ELEC5305 project: Deep Learning-Based Audio Quality Restoration And Enhancement for Remote Speech Transmission

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Abstract

Long-distance speech transmission is easily degraded by noise, echo, packet loss, and channel distortion, often resulting in speech that is difficult to understand. To address these issues, we propose a two-stage system that restores corrupted speech and enhances its spatial listening quality.

Module 1 performs single-channel speech restoration using a lightweight Transformer-based model. By simulating realistic remote-channel impairments and training the network in the STFT domain with spectral and time-domain objectives, the system effectively improves clarity, intelligibility, and perceptual naturalness of degraded speech.

Module 2 then spatializes the restored signal by combining neural ILD prediction with classical DSP methods for ITD and IPD estimation, including GCC-PHAT and phase regression. This hybrid design produces physically consistent and natural-sounding stereo output without requiring the network to learn full complex phase information.

Overall, the proposed framework not only recovers long-distance degraded speech but also delivers an enhanced, immersive stereo experience suitable for communication, conferencing, and multimedia applications.

1. Introduction

Remote speech transmission is highly vulnerable to a wide range of degradations, including background noise[1] [2], packet loss[3], echo[4], codec artifacts, and channel distortion[5]. Speech affected by these impairments often becomes muffled, distorted, or difficult to understand, which not only disrupts everyday communication but also poses challenges for remote education, telemedicine, and collaborative work. As modern communication increasingly relies on long-distance, cross-platform, and real-time voice transmission, restoring degraded speech and improving its intelligibility and naturalness has become an essential research goal.

In this context, this work aims to develop a comprehensive system for restoring and spatially enhancing speech that has been degraded during long-distance transmission. The primary objective is to recover a more intelligible, perceptually natural signal from severely corrupted mono speech, making it closer to the original clean recording while further enhancing the listening experience through spatialization. Unlike conventional speech-enhancement approaches that focus solely on denoising, our system simultaneously targets speech clarity, intelligibility, perceptual quality, and immersive spatial realism, making it suitable not only for communication systems but also for multimedia and interactive applications.

To achieve this, we propose a two-stage deep learning framework consisting of Module 1 and Module 2. Module 1 performs speech restoration using a lightweight convolutional autoencoder and Transformer/Conformer-based temporal modeling[6] [7] [8]. Operating in the

STFT domain, the model predicts spectral masks to reconstruct speech corrupted by realistic impairments such as background noise, bandwidth limitations, echo, and packet loss. By combining frequency-domain and time-domain losses, Module 1 effectively enhances intelligibility and perceptual quality, producing a signal that closely resembles the original clean speech.

Module 2 builds upon the enhanced mono output by introducing spatialization, transforming the restored signal into a natural and immersive stereo representation[9] [10]. It employs a hybrid “neural + classical signal processing” strategy: the neural network predicts interaural level differences (ILD), while interaural time and phase differences (ITD/IPD) are generated through traditional DSP components such as GCC-PHAT, low-frequency phase regression, temporal smoothing, and energy protection mechanisms. This design ensures that the resulting stereo output is perceptually stable, physically coherent, and consistently natural across different acoustic conditions.

Overall, the proposed restoration-and-spatialization system not only revives degraded long-distance speech to improve clarity and naturalness but also adds a realistic stereo spatial experience on top of the enhanced signal. This joint design demonstrates the potential of deep learning to address modern speech transmission challenges and offers a promising foundation for future applications in remote communication, immersive media, and virtual interaction.

2. Research Question

How can a deep learning–based two-stage framework effectively restore speech degraded by long-distance transmission while also reconstructing perceptually coherent stereo spatial cues? This question is particularly important because modern remote communication systems face simultaneous challenges of severe signal degradation and the lack of spatial information, yet most existing methods address only one of these aspects in isolation.

3. Literature Review

In recent years, deep learning has achieved remarkable progress in speech enhancement and spatial audio processing, offering new technical foundations for restoring speech quality under long-distance transmission conditions. Traditional single-channel enhancement methods operating in the time or frequency domain have shown solid performance in suppressing noise, reverberation, and bandwidth-induced distortions. However, their restoration capability remains limited when confronted with the complex and composite degradations commonly observed in remote communication scenarios, such as bandwidth limitations, codec artifacts, echo, packet loss, and mixtures of non-stationary noise. Consequently, deep neural network–based fusion models have emerged as the current state of the art. FullSubNet is a representative example: by combining full-band and sub-band modeling, it captures both global spectral structure and local frequency dynamics, achieving notable improvements in intelligibility and perceptual quality in real-world noisy environments[11].

In parallel, deep learning has also been widely applied to the restoration of packet-loss and channel-induced distortions in remote speech transmission. Within the Packet Loss Concealment (PLC) domain, neural generative approaches have begun to replace traditional repetition- or interpolation-based techniques. The mini-survey on deep PLC highlights that generative adversarial models, autoregressive networks, sequence-to-sequence models, and autoencoders are capable of reconstructing more natural and continuous speech segments, providing a promising direction for addressing severe channel degradations[12]. Among these, Amazon's real-time PLC model proposes a hybrid predictive–generative architecture that leverages a conditioning network together with LPCNet to produce seamless speech segments during loss events, significantly improving robustness in real-time VoIP communication systems[13].

Beyond quality restoration, recent research has also increasingly emphasized the reconstruction of spatial auditory cues in single-channel signals. To convert mono speech into binaural audio that conveys spatial localization, several works explore the ability of neural networks to infer interaural level differences (ILDs) and interaural time/phase differences (ITD/IPD). "Neural Synthesis of Binaural Speech" introduces a position-conditioned framework employing Neural Time Warping to generate direction-dependent binaural waveforms, thereby overcoming limitations inherent to traditional linear DSP-based spatialization methods[14]. Similarly, "Beyond Mono to Binaural" utilizes RGB video and depth maps as auxiliary cues and employs hierarchical cross-modal attention to implicitly capture scene geometry and sound source location, enabling the synthesis of immersive binaural audio consistent with the visual environment[15]. These models share a common assumption: the mono input is clean, and the task centers on recovering spatial cues rather than addressing signal degradations encountered during transmission.

A review of existing literature reveals several notable gaps. First, most prior studies address only a single type of degradation—such as noise suppression, bandwidth extension, or spatialization—and few models can simultaneously handle the combined degradations typical of long-distance speech transmission. Second, binaural spatialization frameworks often rely on external modalities such as vision or geometry, while lightweight, audio-only spatial restoration methods tailored to communication scenarios remain underexplored. Third, existing research on PLC, bandwidth extension, enhancement, and spatial audio largely advances in isolation, with limited efforts to integrate speech restoration and spatial reconstruction into a unified processing pipeline. Motivated by these gaps, the present work proposes a two-stage deep learning framework capable of restoring severely degraded single-channel speech and subsequently generating a perceptually coherent stereo representation, thereby addressing the unmet need for both quality and spatial fidelity in remote speech transmission.

4. Data Source

We use the VoiceBank-DEMAND dataset hosted by the University of Edinburgh's DataShare (link: https://datashare.ed.ac.uk/handle/10283/2791) [16]. The dataset consists of clean speech recordings from the VoiceBank corpus and a variety of real environmental noises from DEMAND, and it is widely adopted in speech enhancement and restoration research. In this work, only the clean speech recordings are used as the content reference. Following standard practice, we adopt the official train–test speaker split to evaluate the model's generalization to unseen speakers. All audio is originally recorded at 48 kHz; we resample it to 16 kHz, downmix stereo recordings to mono, and segment or zero-pad each utterance to a fixed duration (e.g., 4 seconds) to meet the requirements of model training.

Beyond the original dataset structure, we employ a custom RemoteChannelAugmentor to synthesize "remote-transmission–degraded" noisy speech on the fly. Given a clean input waveform, the augmentor stochastically applies several distortion processes commonly observed in real communication channels, including bandwidth compression and bandlimiting (low-rate codec simulation and 300–3.8 kHz telephone-band limiting), colored background noise and "speechbleed"-type leakage noise, echo and residual reverberation modeled via multi-tap FIR responses, as well as frame-level packet loss and burst loss with zero-filling or "last-frame-hold" packet-loss concealment (PLC). The SNR range, loss rates, burst probabilities, and echo decay parameters are randomly drawn from predefined intervals, ensuring that the synthesized degradations span a wide spectrum from mild to severe. This design allows the training process to better reflect real-world remote transmission conditions and provides a unified degraded input for downstream speech restoration and spatialization modules.

5. Methodology

5.1. Overview of the Two-Stage Framework

The proposed speech processing system adopts a two-stage pipeline designed to restore speech degraded during remote transmission and subsequently generate stereo audio with enhanced spatial immersion. The framework consists of a speech restoration stage and a spatialization stage, connected sequentially. The system input is a single-channel speech signal subjected to various remote-channel degradations, including codec-induced compression, bandwidth limitation, colored background noise, echo, residual reverberation, and frame-level packet loss. These distortions are introduced during training using a custom RemoteChannelAugmentor, which stochastically superimposes multiple real-world communication impairments, enabling the model to learn robust restoration across diverse degradation conditions.

In the first stage, a lightweight Transformer-based model performs enhancement and restoration of the degraded speech. The noisy waveform is converted into STFT magnitude features, and the model predicts a spectral mask to recover a clean target magnitude, followed by ISTFT reconstruction to obtain an enhanced single-channel waveform. The restored speech not only exhibits reduced noise and improved intelligibility but also maintains a time–frequency representation fully compatible with the subsequent spatialization stage, ensuring a coherent transition between stages.

The second stage takes the cleaned and repaired high-quality mono speech as input and constructs a spatially coherent stereo field. Instead of relying on large end-to-end stereo generation networks, this work adopts a hybrid strategy that combines neural ILD prediction with classical signal-processing methods for ITD and IPD estimation. Specifically, the model predicts magnitude masks for the left and right channels in the STFT domain, which determine the interaural level difference (ILD)[17]. Meanwhile, the interaural time and phase differences (ITD/IPD)[18] [19] are derived from a stereo teacher reference using established DSP techniques such as GCC-PHAT, low-frequency phase regression, and EMA smoothing. The predicted ILD and analytically estimated ITD/IPD are jointly applied to the restored STFT representation, after which ISTFT reconstruction yields a natural, stable, and physically consistent stereo signal.

The two-stage framework is unified through a shared STFT-based representation, allowing seamless integration between speech restoration and spatialization. Although the two stages are independently trainable and analytically distinguishable, they remain tightly aligned in terms of feature format and data flow. As a result, the system progressively transforms real-world degraded remote speech into a clear, natural, and spatially immersive stereo output.

5.2. Reasons for the choice of methods

To address the central research question of this study—how to simultaneously achieve speech restoration and spatial enhancement under severe degradations induced by long-distance transmission—this work adopts a two-stage deep learning framework consisting of a lightweight

Transformer-based restoration module and a neural–DSP hybrid spatialization module. The selection of this approach is grounded in clear technical motivations and evidence from prior literature. First, compared with traditional convolutional autoencoders or GAN-based enhancement models, lightweight Transformer/Conformer architectures offer superior capability in modeling long-range temporal–spectral dependencies and capturing global structures in speech signals. These properties are particularly advantageous for handling complex and non-stationary degradations commonly found in remote transmission scenarios, such as bandwidth constraints, codec distortions, and echo artifacts. FullSubNet, for example, has demonstrated the effectiveness of full-band and sub-band spectral modeling in enhancing intelligibility under real-world noisy conditions [11].

Second, for spatial enhancement, this study adopts a hybrid strategy that combines neural ILD prediction with traditional DSP-based ITD/IPD reconstruction, rather than relying on fully end-to-end binaural generation methods. While state-of-the-art systems such as *Neural Synthesis of Binaural Speech* [14] and *Beyond Mono to Binaural* [15] are capable of producing high-quality binaural audio, they typically depend on auxiliary visual modalities (e.g., depth maps) or strong spatial priors, and often assume that the mono input signal is clean or only mildly degraded. These assumptions do not hold in real long-distance communication, where speech is severely corrupted by bandwidth limitations, mixtures of non-stationary noise, packet loss, and channel distortions. Consequently, this work adopts a more communication-oriented design: the neural network predicts interaural level differences (ILDs), while GCC-PHAT and low-frequency phase regression are used to generate physically coherent interaural time and phase differences (ITD/IPD). This hybrid approach preserves physical interpretability, enhances robustness across acoustic conditions, and avoids the instability often observed in fully learned phase-generation systems.

Furthermore, compared with end-to-end spatialization approaches, the hybrid design significantly reduces model complexity and computational cost, making it more suitable for real-time deployment in communication applications. For these reasons, the proposed two-stage architecture is well aligned with the functional demands of remote speech transmission and provides a stable, interpretable, and efficient solution capable of addressing both speech restoration and spatial enhancement tasks simultaneously.

5.3. New Tools and Techniques Learned

During the implementation of the proposed framework, several new tools and techniques were acquired to support both speech restoration and spatialization. On the restoration side, the study learned to use PyTorch's STFT/ISTFT modules to manipulate spectral representations, perform magnitude–phase decomposition, and apply time–frequency masking with stable gradient propagation. The work also involved learning residual MLP structures and lightweight Transformer attention mechanisms for modeling complex non-stationary degradations.

For spatialization, the study gained experience with GCC-PHAT for robust interaural time-delay estimation, as well as DSP techniques such as low-frequency phase regression, IPD coherence constraints, and energy smoothing to ensure stable binaural phase reconstruction. Additionally, perceptual weighting methods including A-weighting and ERB-band smoothing were adopted to better align training with human auditory sensitivity.

Finally, the work involved learning packet-loss simulation techniques commonly used in remote communication research, including random loss masking, burst-loss modeling, and basic PLC-style reconstruction pipelines. These tools allowed the training data to more accurately reflect real VoIP transmission conditions. Overall, these newly learned techniques significantly improved the feasibility and robustness of the proposed system.

5.4. Data Preparation and Remote-Channel Simulation

In the data preparation stage, our pipeline jointly serves two tasks: (i) constructing paired noisy–clean training samples for the speech restoration module, and (ii) establishing the mapping between non-spatial mono signals and target stereo signals for the spatialization module. We build our training and test corpora on the VoiceBank–DEMAND dataset, which contains clean speech from multiple speakers (VoiceBank) and noise recordings from diverse real-world environments (DEMAND) and is widely used in speech enhancement research. In our experiments, only the clean speech is used as the linguistic content reference. All utterances are resampled to 16 kHz, converted to mono, and normalized via the `read_mono_resample` function into waveforms $\mathbf{x} \in [-1, 1]^T$. If the original duration is shorter than a fixed length $T = \text{SEGMENT\_SAM}$, zero-padding is applied; if it is longer than T, a random segment of length T is cropped. This procedure improves the robustness and generalization of the model to different speakers, contents, and acoustic conditions.

To emulate typical degradations encountered in real-world remote speech transmission, we design a custom *RemoteChannelAugmentor* that randomly applies multiple communication-channel distortions to clean speech during training. The simulated factors include bandwidth limitation and coding artifacts (e.g., 300–3800 Hz band-limiting and low-bitrate codec aliasing), colored background noise and speechbleed-type interfering speech, echo and residual reverberation modeled by multi-tap FIR responses, as well as frame-level random and burst packet loss combined with packet-loss concealment strategies based on zero-filling or previous-frame holding. All parameters are sampled from predefined ranges so that the training set covers a continuum of channel conditions from mildly compressed to severely degraded, closely approximating scenarios such as remote teaching, online meetings, and VoIP calls.

On top of this, we further apply *perceptual enhancement* to the clean speech in order to construct reference targets that better match human auditory preference. First, ENHANCER_MONO operates in the STFT domain, applying A-weighting to the magnitude spectrum and boosting energy in the high-frequency presence region, while ERB-band smoothing is used to suppress spectral roughness. This jointly improves the perceived clarity, brightness, and overall smoothness of the speech. Then, short-tailed synthetic early reflections are introduced in the time domain, followed by A-weighted loudness calibration to ensure that the enhanced mono signal exhibits a moderate perceptual lift while preserving natural timbre. The resulting enhanced clean signal can be used either as the training target for Module 1 (noisy–clean learning) or as a refined input reference for the subsequent spatialization stage.

To obtain device-consistent stereo references with stable spatial impression, we employ ENHANCER_STEREO to expand the ENHANCER_MONO-processed mono signal into a stereo teacher with explicit spatial attributes. Concretely, a controllable interaural time difference (ITD) on the order of milliseconds is injected between the left and right channels, high-frequency shelving filters with opposite gains are applied to the two channels to strengthen interaural level differences (ILD), and mild early reflections together with late-decay tails are added to emulate spatial diffusion in real acoustic environments. The resulting stereo teacher not only serves as the supervision target for Module 2 (mono-to-stereo spatial generation), but also provides a unified and stable reference for subsequent objective spatial metrics such as ITD, ILD, and IACC.

5.4.1. Module 1: Noisy–Clean Speech Pair Construction

For the speech restoration module, the training data consist of paired samples $(x_{\text{noisy}}, x_{\text{clean}})$, where x_{clean} is derived from the clean portion of the VoiceBank–DEMAND corpus, and x_{noisy} is generated via a custom RemoteChannelAugmentor. This augmentor simulates remote-channel degradations directly in the waveform domain. For each utterance, the system probabilistically chooses whether to use an existing "noisy packet" recording or synthesize a degraded version. The latter is adopted in most experimental settings to allow fine-grained control over distortion types and statistical distributions.

The augmentor maps clean speech $x(t)$ to its degraded form:

$$y = \mathcal{A}(x; \theta) = \mathcal{P}\big(\mathcal{N}(\mathcal{E}(\mathcal{B}(\mathcal{C}(x))))\big),$$

where $\mathcal{C}$ denotes codec/resampling distortion, $\mathcal{B}$ bandwidth limitation, $\mathcal{E}$ echo/reverberation, $\mathcal{N}$ colored noise, and $\mathcal{P}$ packet loss concealment (PLC).

The clean signal first undergoes a resampling process from the target rate of 16 kHz down to 8 kHz and back, which introduces aliasing and bandwidth folding effects similar to those found in low-rate codec transmission. A subsequent μ-law companding stage further adds nonlinear compression artifacts commonly observed in telecommunication channels. Formally, the companding and expansion operations are defined as:

$$\tilde{x} = \operatorname{sgn}(x)\frac{\ln(1+\mu|x|)}{\ln(1+\mu)}, \qquad x_\mu = \operatorname{sgn}(\tilde{x})\frac{1}{\mu}\left(\exp(\ln(1+\mu)|\tilde{x}|) - 1\right),$$

with $\mu = 255$. These processing steps jointly emulate the quantization, bandwidth reduction, and reconstruction behavior of real-world speech transmission systems.

To emulate the restricted bandwidth of traditional telecommunication channels, a spectral trimming operation is applied in the STFT domain. Given the time–frequency representation

$$X(f,k) = \text{STFT}\{x(t)\},$$

a binary frequency mask $M(f) \in \{0,1\}$ removes components outside the desired passband, leading to

$$Y(f,k) = M(f)X(f,k), \qquad y(t) = \text{ISTFT}\{Y(f,k)\}.$$

This procedure replicates the perceptual dullness and high-frequency attenuation typical of narrowband communication systems such as the 300–3400 Hz telephone band.

Early reflections and room-like reverberation are simulated using a finite-length impulse response. An impulse response $h[n]$ is constructed with a direct path and several delayed taps,

$$h[0] = 1, \qquad h[d_i] = a_i, \quad i = 1, \dots, N_{\text{tap}},$$

where each delay d_i typically corresponds to tens of milliseconds and the amplitudes a_i model the reflection strength. The degraded signal is obtained by linear convolution,

$$y_{\text{echo}}[n] = (x * h)[n],$$

thereby producing a mixture of direct sound and multi-path reflections.

Colored background noise is generated through a first-order autoregressive filter,

$$z[n] = a\, z[n-1] + (1-a)\, w[n],$$

where $w[n] \sim \mathcal{N}(0,1)$ and $a \approx 0.985$ controls the spectral tilt, producing noise ranging from white to brown characteristics. To simulate "speechbleed" conditions, segments of the same utterance may be randomly shuffled and mixed with the colored noise prior to scaling. Given a target SNR interval, the corresponding noise power is computed as

$$P_n^* = \frac{P_x}{10^{\mathrm{SNR_{dB}}/10}},$$

and the final degraded signal is obtained as

$$y_{\text{noisy}} = x + \alpha n,$$

where α normalizes the noise to meet the desired power level.

Packet-loss effects are introduced at the frame level, where segments of length $L = \lfloor f_s \cdot 20 \text{ ms} \rfloor$ are randomly removed according to a loss probability p_{loss}. Each dropped frame is subsequently reconstructed using either zero-filling or a hold-mode packet loss concealment (PLC) strategy, after which the waveform is amplitude-clipped to maintain numerical stability. In parallel, a perceptual enhancement process is applied to the clean speech to generate a more realistic reference target. This procedure incorporates A-weighting to emphasize perceptually important frequency regions, ERB-scale smoothing to suppress spectral roughness, and a presence-boosting filter to strengthen high-frequency clarity. Formally, the enhanced magnitude representation is computed as

$$\tilde{M}(f,k) = \text{Smooth}_{\text{ERB}}\big(A(f) \cdot P(f) \cdot |X(f,k)|\big),$$

providing a refined perceptual reference that guides both the restoration and spatialization stages.

5.4.2. Module 2: Mono–Stereo Spatial Pair Construction

The spatialization module converts high-quality mono speech into stereo. The data pairs $(x_{\text{in}}, s_{\text{st}})$ are constructed where x_{in} is non-spatial mono (either clean or perceptually enhanced) and $s_{\text{st}} = [L(t), R(t)]^\top$ is produced by ENHANCER STEREO:

$$s_{\text{st}} = \mathcal{T}(x_{\text{clean}}).$$

ITD and ILD cues are introduced via:

$$L(t) = y(t), \qquad R(t) = y(t - \Delta t),$$

with $\Delta t \approx 0.3$ ms. High-frequency shelving (±stereo_shelf_db) adds ILD cues, and early reflections are simulated:

$$L'(t) = L(t) + \sum_i g_i L(t - d_i), \qquad R'(t) = R(t) + \sum_i g_i R(t - d_i - 2 \text{ ms}).$$

Noise tails $\eta_{L,R}(t)$ are added:

$$s_{\text{st}}(t) = \big[L'(t) + \eta_L(t),\ R'(t) + \eta_R(t)\big]^\top.$$

A CleanToStereoTeacherDataset generates stereo targets while maintaining the same segment length $T = SEGMENT\_SAM$ as Module 1. Evaluation uses full-length sequences. Module 2 predicts ILD masks in the STFT domain, whereas ITD/IPD cues are inferred using GCC-PHAT and low-frequency phase regression.

Consistent preprocessing (sampling rate, segmentation, and STFT parameters) ensures unified evaluation for both perceptual and spatial metrics.

5.5. Stage 1: Remote Speech Restoration Network

The objective of Stage 1 is to restore the quality of single-channel speech after remote-channel degradation, producing a high-fidelity signal that serves as the reference for subsequent spatialization and objective evaluation. In this stage we focus solely on content and quality restoration rather than spatial cues, so both the input and output are mono waveforms. Given a degraded input $x_{\text{noisy}}(t)$ and its corresponding clean reference $x_{\text{clean}}(t)$, Stage 1 learns a mask–prediction function

$$\mathcal{F}_\theta : \ \log\bigl(1 + |X_{\text{noisy}}(f,k)|\bigr) \ \longmapsto \ \hat{M}(f,k), \tag{1}$$

where $X_{\text{noisy}}(f,k) = \text{STFT}\{x_{\text{noisy}}(t)\}$ is the short-time Fourier transform (STFT) and $\hat{M}(f,k)$ is the real-valued magnitude mask predicted by the network. The estimated magnitude spectrum is obtained as

$$\hat{X}(f,k) = \hat{M}(f,k)\,|X_{\text{noisy}}(f,k)|, \qquad \hat{x}(t) = \text{ISTFT}\left(\hat{X}(f,k)\, e^{j\angle X_{\text{noisy}}(f,k)}\right), \tag{2}$$

where the phase is directly taken from the noisy signal to reduce model complexity and stabilize training.

5.5.1. Model Architecture

Stage 1 adopts a lightweight Transformer encoder as the backbone, combined with temporal depthwise convolutions and a frequency–control interpolation scheme to efficiently model the time–frequency plane. The input waveform $x_{\text{noisy}}(t)$ is first transformed by the STFT into a complex spectrum $X_{\text{noisy}}(f,k)$. Its magnitude is compressed via $\log(1+\cdot)$,

$$S(f,k) = \log\bigl(1 + |X_{\text{noisy}}(f,k)|\bigr), \tag{3}$$

and reshaped into a tensor of size $[B,1,F,T]$, where $F = N_{\text{FFT}}/2+1$ is the number of frequency bins and T is the number of time frames.

For each frame, a linear projection maps the F-dimensional spectral feature into a d_{model}-dimensional latent vector:

$$\mathbf{z}_k = \mathbf{W}_{\text{in}}\mathbf{s}_k + \mathbf{b}_{\text{in}}, \quad \mathbf{s}_k \in \mathbb{R}^F, \ \mathbf{z}_k \in \mathbb{R}^{d_{\text{model}}}, \tag{4}$$

yielding a temporal sequence $\{\mathbf{z}_k\}_{k=1}^T$. A one-dimensional depthwise convolution along the time axis is then applied to capture local temporal correlations and enhance expressiveness. A sinusoidal positional encoding $\mathbf{p}_k$ is added to obtain

$$\tilde{\mathbf{z}}_k = \mathbf{z}_k + \mathbf{p}_k. \tag{5}$$

The core encoder consists of several Transformer encoder layers, each including multi-head self-attention and a feedforward network with a pre-norm layout and GELU activation:

$$\mathbf{h}_k = \text{TransformerEncoder}\left(\tilde{\mathbf{z}}_{1:T}\right)_k, \qquad k = 1,\ldots,T. \tag{6}$$

Instead of regressing a mask value for every frequency bin directly, the network defines K "control bands" $\{f_j^{\text{ctrl}}\}_{j=1}^K$ along the frequency axis. A linear layer outputs the corresponding control values for each frame:

$$\mathbf{u}_k = \text{softplus}(\mathbf{W}_{\text{ctrl}}\mathbf{h}_k + \mathbf{b}_{\text{ctrl}}) \in \mathbb{R}^K. \tag{7}$$

A precomputed interpolation matrix $\mathbf{W}_{\text{interp}} \in \mathbb{R}^{F\times K}$ then smoothly expands these K control points to all F frequency bins:

$$\hat{\mathbf{m}}_k = \mathbf{W}_{\text{interp}}\mathbf{u}_k \in \mathbb{R}^F. \tag{8}$$

The resulting three-dimensional mask tensor $\hat{M}(f,k)$ is constrained by a softplus activation and an upper bound (e.g., 1.5) to ensure non-negativity and to prevent excessive amplification. In Stage 1 the optional phase head is disabled and only magnitude masks are predicted, keeping the parameter count in the range of roughly 0.1–0.2 million and making the model suitable as a lightweight remote speech restoration subnetwork.

5.5.2. Loss Functions

To simultaneously account for spectral fidelity, time-domain alignment, and perceived quality, Stage 1 employs a multi-component loss function. Let $\hat{x}(t)$ denote the restored waveform, $\hat{X}(f,k)$ its magnitude spectrum, and $X_{\text{clean}}(f,k)$ the magnitude spectrum of the clean reference. The primary spectral term is a log-magnitude mean squared error (MSE):

$$\mathcal{L}_{\text{spec}} = \left\| \log\big(1+|\hat{X}(f,k)|\big) - \log\big(1+|X_{\text{clean}}(f,k)|\big) \right\|_2^2, \tag{9}$$

which directly constrains the distribution of spectral energy and facilitates the recovery of formants and high-frequency details.

Because the human auditory system is also sensitive to waveform discrepancies, an L1 loss is applied to the time-domain signals after alignment:

$$\mathcal{L}_{\text{time}} = \frac{1}{L}\sum_{t=1}^{L} \left|\hat{x}(t) - x_{\text{clean}}(t)\right|, \tag{10}$$

where L is the minimum length after truncation.

To quantify overall quality and distortion, the third component is the scale-invariant signal-to-distortion ratio (SI-SDR). Let $\tilde{s}$ and $\tilde{x}$ denote the mean-removed reference and estimated signals, respectively. Define

$$\alpha = \frac{\langle \tilde{x}, \tilde{s}\rangle}{\|\tilde{s}\|_2^2}, \quad \hat{s} = \alpha\tilde{s}, \quad e = \tilde{x} - \hat{s}, \tag{11}$$

then

$$\text{SI-SDR}(\tilde{x},\tilde{s}) = 10\log_{10}\frac{\|\hat{s}\|_2^2}{\|e\|_2^2}. \tag{12}$$

During training we use its negative, $\mathcal{L}_{\text{SI-SDR}} = -\,\text{SI-SDR}$, as a loss term to encourage improvements in global speech quality.

To avoid overly fluctuating masks in the time–frequency domain, Stage 1 further introduces a smoothness and sparsity regularizer. With $\hat{M}(f,k)$ denoting the predicted mask, we define

$$\mathcal{L}_{\text{mask}} = w_t\big\|\hat{M}(f,k) - \hat{M}(f,k-1)\big\|_1 + w_f\big\|\hat{M}(f,k) - \hat{M}(f-1,k)\big\|_1 + w_1\|\hat{M}(f,k)\|_1, \tag{13}$$

where w_t, w_f, and w_1 control temporal smoothness, spectral smoothness, and overall L1 regularization, respectively. This term suppresses excessively sharp mask structures and helps mitigate "musical noise" artifacts.

Combining all components, the overall loss for Stage 1 is

$$\mathcal{L} = \lambda_{\text{spec}}\mathcal{L}_{\text{spec}} + \lambda_{\text{time}}\mathcal{L}_{\text{time}} + \lambda_{\text{SI-SDR}}\mathcal{L}_{\text{SI-SDR}} + \lambda_{\text{mask}}\mathcal{L}_{\text{mask}}, \tag{14}$$

where λ_{spec}, λ_{time}, $\lambda_{\text{SI-SDR}}$, and λ_{mask} are empirically chosen weights. In the current implementation, the spectral term dominates ($\lambda_{\text{spec}} \approx 1$), while the time-domain L1 and SI-SDR terms contribute with smaller coefficients, and the mask regularizer accounts for a modest fraction of the total loss.

5.5.3. Training Strategy and Evaluation

In terms of optimization, Stage 1 uses the Adam optimizer to train all parameters end-to-end. Automatic mixed precision (AMP) is employed to reduce memory usage and accelerate training. The initial learning rate is set to 5×10^{-4}, and can be combined with cosine or step-wise learning-rate schedulers. All STFT and ISTFT operations are implemented in PyTorch to maintain differentiability; for stability, however, the noisy magnitude $|X_{\text{noisy}}|$ is detached during the forward pass so that gradients do not propagate back into the front-end transform. Before each parameter update, gradient norms are clipped (e.g., $\|\nabla\theta\|_2 \leq 5$) to prevent sporadic explosions.

To monitor generalization on unseen data, we evaluate the log-spectral MSE $\mathcal{L}_{\text{spec}}$ and SI-SDR on a held-out validation set, tracking their evolution across epochs to select the best model checkpoint. Furthermore, for a given test utterance, the system computes a richer set of objective metrics, including SI-SDR[20], segmental SNR[2], log-spectral distance (LSD), mel-cepstral distortion (MCD)[21], mel-spectral LSD (mel-LSD), and spectral convergence. Waveforms, error signals, and log-power spectrograms of the noisy, denoised, and clean signals are visualized side-by-side.

5.6. Stage 2: Neural–Classical Hybrid Spatialization

Stage 2 converts the clean mono speech into a binaural stereo signal, forming a two-stage pipeline of "remote enhancement + local spatialization". The training data consist of clean speech $x_{\text{clean}}[n]$ and its corresponding stereo teacher $\mathbf{y}_{\text{teach}}[n] = [y_L[n], y_R[n]]^\top$, where the teacher is generated offline by the perceptual enhancement module ENHANCER_STEREO without any additional remote-channel degradation or noise. Stage 2 therefore learns a mapping

$$x_{\text{in}}[n] \longrightarrow \mathbf{y}_{\text{stereo}}[n], \tag{15}$$

where $x_{\text{in}}[n]$ is the (denoised or perceptually enhanced) mono input, and $\mathbf{y}_{\text{stereo}}[n]$ is the desired stereo output.

In the proposed design, the neural network is responsible only for predicting left/right magnitude masks, which encode interaural level difference (ILD), whereas interaural time and phase differences (ITD/IPD) are estimated entirely by classical DSP methods such as GCC-PHAT and low-frequency phase regression. This leads to a "neural amplitude + classical phase" hybrid spatialization framework that combines data-driven modeling of energy distribution with physically interpretable time/phase cues.

5.6.1. ILD Prediction Network

Given the mono input signal $x_{\text{in}}[n]$, we first compute its short-time Fourier transform (STFT)

$$X(f,k) = \text{STFT}\{x_{\text{in}}[n]\}, \tag{16}$$

and obtain its magnitude and phase: $|X(f,k)|$ and $\angle X(f,k)$. As in Stage 1, the magnitude is compressed using a log transform,

$$S(f,k) = \log\bigl(1 + |X(f,k)|\bigr), \tag{17}$$

and stacked into a tensor of shape $[B, 1, F, T]$ as the network input.

The spatialization network adopts a frame-wise residual fully connected architecture (ResFC / frame-wise MLP). For each time frame k, the frequency-domain vector $\mathbf{s}_k \in \mathbb{R}^F$ is first normalized at the frame level and linearly projected into a hidden space:

$$\mathbf{h}_k^{(0)} = \mathrm{LN}(\mathbf{s}_k), \qquad \mathbf{z}_k^{(0)} = W_{\mathrm{in}}\mathbf{h}_k^{(0)} + \mathbf{b}_{\mathrm{in}}. \tag{18}$$

Then, $\mathbf{z}_k^{(0)}$ is passed through several residual MLP blocks:

$$\mathbf{z}_k^{(\ell+1)} = \mathbf{z}_k^{(\ell)} + \mathrm{MLP}_\ell\Big(\mathrm{LN}\Big(\mathbf{z}_k^{(\ell)}\Big)\Big), \tag{19}$$

where each MLP_ℓ is a two-layer feed-forward network with GELU activation and optional dropout. This design preserves the advantages of residual connections and layer normalization, while avoiding the high computational overhead of self-attention, and is well suited for lightweight modeling of frame-wise spectral vectors.

Finally, the hidden representation $\mathbf{z}_k^{(L)}$ is mapped to the left/right magnitude masks:

$$\tilde{\mathbf{m}}_k = W_{\mathrm{mask}}\mathbf{z}_k^{(L)} + \mathbf{b}_{\mathrm{mask}} \in \mathbb{R}^{2F}, \tag{20}$$

$$\mathbf{m}_c(f,k) = \mathrm{clip}_{[0,m_{\max}]}\big(\mathrm{softplus}(\tilde{\mathbf{m}}_c(f,k))\big), \quad c \in \{L, R\}, \tag{21}$$

where $\mathrm{softplus}(\cdot)$ enforces non-negativity and $\mathrm{clip}_{[0,m_{\max}]}$ prevents extreme amplification in any individual band. The predicted magnitude spectra for the two channels are then

$$|Y_c(f,k)| = \mathbf{m}_c(f,k)\,|X(f,k)|, \quad c \in \{L, R\}. \tag{22}$$

From an acoustic standpoint, $\mathbf{m}_L$ and $\mathbf{m}_R$ jointly encode the energy ratio between the two ears in each frequency band, i.e., a data-driven pattern of interaural level differences (ILDs)[17].

5.6.2. Classical ITD/IPD Estimation

In contrast to the magnitude branch, Stage 2 does not use a neural network to directly estimate phase differences. Instead, ITD and IPD are obtained from the stereo teacher using well-established signal processing methods, and then combined with the predicted magnitudes. This yields a stable and physically interpretable spatial phase structure.

Given the teacher stereo signal $\mathbf{y}_{\mathrm{teach}}[n]$ with channels $y_L[n]$ and $y_R[n]$, we compute their STFTs:

$$Y_L(f,k),\ Y_R(f,k) = \mathrm{STFT}\{y_L[n]\},\ \mathrm{STFT}\{y_R[n]\}. \tag{23}$$

Using GCC-PHAT (generalized cross-correlation with phase transform), we construct the cross-power spectrum

$$G_{LR}(f,k) = \frac{Y_L(f,k)\,Y_R^*(f,k)}{|Y_L(f,k)\,Y_R(f,k)| + \varepsilon}, \tag{24}$$

and obtain the cross-correlation function

$$g_{LR}(\tau,k) = \mathcal{F}^{-1}_{f\to\tau}\{G_{LR}(f,k)\}. \tag{25}$$

Within a physiologically plausible delay range $|\tau| \le \tau_{\max}$, the ITD estimate and the interaural cross-correlation (IACC) are given by

$$\hat{\tau}_{\mathrm{GCC}}(k) = \arg\max_{|\tau|\le\tau_{\max}} |g_{LR}(\tau,k)|, \quad \mathrm{IACC}(k) = \max_{|\tau|\le\tau_{\max}} |g_{LR}(\tau,k)|. \tag{26}$$

To further exploit low-frequency IPD information, the phase difference

$$\phi_{\mathrm{IPD}}(f,k) = \angle Y_L(f,k) - \angle Y_R(f,k) \tag{27}$$

in the band $|f| \le f_{\max}$ is approximated by a linear model $\phi_{\mathrm{IPD}}(f,k) \approx \omega_f \, \tau_{\mathrm{reg}}(k)$ with $\omega_f = 2\pi f$. A weighted least-squares fit, using amplitude-based weights to emphasize high-energy bands, yields a regression-based ITD $\tau_{\mathrm{reg}}(k)$. Frame-wise energy gating and an exponential moving average (EMA) are then applied to suppress jitter in low-energy frames. The final ITD estimate combines the GCC-PHAT and regression results:

$$\hat{\tau}(k) = (1-\beta)\, \tau_{\mathrm{reg}}(k) + \beta \, \hat{\tau}_{\mathrm{GCC}}(k), \quad \beta \in [0,1]. \tag{28}$$

The resulting frame-level ITD is converted into a frequency-dependent IPD:

$$\Delta\phi(f,k) = \omega_f \, \hat{\tau}(k), \tag{29}$$

and modulated by a frequency weighting window $w(f)$ that gradually attenuates IPD above a cutoff frequency, reflecting the fact that human ITD sensitivity is strongest at low frequencies while high-frequency spatial perception relies more on ILD. The phase rotators for the two channels are then

$$R_L(f,k) = e^{+j\, w(f) \Delta\phi(f,k)}, \qquad R_R(f,k) = e^{-j\, w(f) \Delta\phi(f,k)}. \tag{30}$$

Finally, the predicted stereo STFT is constructed by combining these phase rotators with the mono phase and the predicted magnitudes:

$$\widehat{Y}_L(f,k) = |Y_L(f,k)| \, e^{j\angle X(f,k)} \, R_L(f,k), \qquad \widehat{Y}_R(f,k) = |Y_R(f,k)| \, e^{j\angle X(f,k)} \, R_R(f,k), \tag{31}$$

and the time-domain signals $\hat{y}_L[n]$ and $\hat{y}_R[n]$ are obtained via inverse STFT. In this pipeline, the neural network learns "how to distribute energy" between the two channels (ILD), while the time/phase differences (ITD/IPD) responsible for spatial localization are provided by explicit physical models[18] [19].

5.6.3. Loss Functions for Spatial Training

Since Stage 2 only trains the magnitude masks, the loss functions focus on spectral fidelity, downmix quality, and frequency-band ILD consistency. Let $Y_L(f,k), Y_R(f,k)$ denote the teacher STFT and $\widehat{Y}_L(f,k), \widehat{Y}_R(f,k)$ the prediction.

To reduce sensitivity to absolute magnitude scaling, both predicted and target spectra are first passed through a $\log(1+\cdot)$ compression before computing their mean squared error. The spectral reconstruction loss is therefore defined as

$$\begin{aligned} \mathcal{L}_{\mathrm{spec}} = & \left\| \log\big(1+|\widehat{Y}_L|\big) - \log\big(1+|Y_L|\big) \right\|_2^2 \\ & + \left\| \log\big(1+|\widehat{Y}_R|\big) - \log\big(1+|Y_R|\big) \right\|_2^2. \end{aligned}$$

This formulation enforces spectral-level similarity between the predicted and teacher magnitudes, while the logarithmic compression stabilizes the loss against large inter-frame amplitude variations and improves training robustness.

The downmix perceptual loss is computed by first collapsing both the predicted and teacher stereo signals to mono through simple averaging:

$$\hat{s}_{\mathrm{dm}}[n] = \tfrac{1}{2}\big(\hat{y}_L[n] + \hat{y}_R[n]\big), \qquad s_{\mathrm{dm}}[n] = \tfrac{1}{2}\big(y_L[n] + y_R[n]\big), \tag{32}$$

after which the scale-invariant signal-to-distortion ratio SI-SDR($\hat{s}_{\mathrm{dm}}, s_{\mathrm{dm}}$) is evaluated. During training, the negative value of this metric is used as the downmix loss $\mathcal{L}_{\mathrm{SI-SDR(dm)}}$, which encourages the downmixed prediction to retain both the perceptual quality and semantic content of the teacher signal. This constraint prevents the spatialization process from introducing overly aggressive inter-channel differences that could degrade timbral fidelity or speech intelligibility.

To explicitly control the interaural energy difference between the two stereo channels, we first compute the time-averaged RMS for each frequency band:

$$r_c(f) = \sqrt{\frac{1}{T}\sum\nolimits_k |Y_c(f,k)|^2}, \qquad \hat{r}_c(f) = \sqrt{\frac{1}{T}\sum\nolimits_k |\widehat{Y}_c(f,k)|^2}, \tag{33}$$

from which the teacher and predicted ILD spectra are obtained:

$$\mathrm{ILD}_{\mathrm{tgt}}(f) = 20\log_{10}\frac{r_L(f)}{r_R(f)}, \qquad \mathrm{ILD}_{\mathrm{pred}}(f) = 20\log_{10}\frac{\hat{r}_L(f)}{\hat{r}_R(f)}. \tag{34}$$

The frequency-band ILD loss is then defined as

$$\mathcal{L}_{\text{ILD-FB}} = \left\|\mathrm{ILD}_{\mathrm{pred}}(f) - \mathrm{ILD}_{\mathrm{tgt}}(f)\right\|_2^2, \tag{35}$$

which compels the predicted ILD profile to match the teacher's ILD distribution across frequency, thereby preserving spatial position, width, and overall stereo localization cues. Combining this loss with the spectral reconstruction and downmix perceptual terms yields the full Stage 2 training objective:

$$\mathcal{L}_{\mathrm{Stage2}} = \lambda_{\mathrm{spec}}\,\mathcal{L}_{\mathrm{spec}} + \lambda_{\text{SI-SDR(dm)}}\,\mathcal{L}_{\text{SI-SDR(dm)}} + \lambda_{\text{ILD-FB}}\,\mathcal{L}_{\text{ILD-FB}}, \tag{36}$$

where the weighting coefficients $\lambda_{\bullet}$ (set to approximately 0.5, 0.2, and 0.9 in our experiments) allow balancing between spectral fidelity, overall audio quality, and spatial consistency.

5.6.4. Evaluation Protocol for Spatialization

To assess the effectiveness of the neural–classical hybrid spatialization, we adopt a set of evaluation metrics that directly reflect spatial cues as well as overall signal quality. All metrics are computed on full-length utterances from the clean–to–stereo teacher dataset, and use the same STFT configuration as in training to ensure consistency.

For each test utterance, frequency-domain spatial metrics are computed over the input mono audio, the predicted stereo output, and the teacher stereo reference. Because mono inputs have identical left and right channels, they exhibit trivial spatial characteristics—zero ILD and IPD, and an IACC of unity. For stereo signals, ILD curves are obtained from band-wise RMS energies following the same formulation used in the ILD frequency-band loss $\mathcal{L}_{\text{ILD-FB}}$. The deviation between predicted and teacher ILD distributions is quantified by the root-mean-square error

$$\mathrm{ILD\_RMSE\_dB} = \sqrt{\mathbb{E}_f\left[(\mathrm{ILD}_{\mathrm{pred}}(f) - \mathrm{ILD}_{\mathrm{tgt}}(f))^2\right]}. \tag{37}$$

Phase-related spatial cues are summarized using interaural phase differences (IPD), and the average absolute IPD deviation is computed as

$$\mathrm{IPD\_L1} = \mathbb{E}_f\left[|\mathrm{IPD}_{\mathrm{pred}}(f) - \mathrm{IPD}_{\mathrm{tgt}}(f)|\right]. \tag{38}$$

Temporal coherence between stereo channels is assessed through frame-wise interaural cross-correlation (IACC). The mean IACC difference relative to the teacher stereo is reported as

$$\Delta\text{IACC}_{\text{mean}} = \mathbb{E}_k\big[\text{IACC}_{\text{pred}}(k) - \text{IACC}_{\text{tgt}}(k)\big], \tag{39}$$

and is further contrasted with the mono baseline, which naturally exhibits minimal IACC variation due to the absence of spatial cues. These metrics jointly characterize the accuracy of spatial reproduction across frequency and time, capturing spatial width, lateralization consistency, and phase alignment quality.

To verify that the spatialization process does not degrade the underlying speech quality, we evaluate the downmix SI-SDR between the predicted downmixed signal $\hat{s}_{\text{dm}}[n]$ and the teacher downmix $s_{\text{dm}}[n]$. This provides a perceptual quality measure that is independent of spatial attributes. In addition to reporting the absolute SI-SDR value, we also compute the improvement over the original mono input,

$$\Delta\text{SI-SDR} = \text{SI-SDR}(\hat{s}_{\text{dm}}, s_{\text{dm}}) - \text{SI-SDR}(x_{\text{in}}, s_{\text{dm}}), \tag{40}$$

which quantifies how much the spatialization stage preserves or enhances the perceptual fidelity relative to the clean mono reference. This metric ensures that the generation of stereo cues does not come at the cost of intelligibility or spectral accuracy in the downmixed signal.

In addition to scalar metrics, we visualize ILD/IPD curves and IACC trajectories, and inspect spectrograms of the input, prediction, and teacher. Subjective listening tests are also conducted to confirm that the predicted stereo signals exhibit stable lateralization, natural spatial width, and minimal coloration.

6. Experiment and Results

6.1. Experimental Setup

All experiments are conducted in a two-stage pipeline that separates speech denoising from spatialization. Audio is resampled to a target sampling rate of 16 kHz, and training is performed on fixed-length segments while evaluation uses full utterances. The training and evaluation data are drawn from the VoiceBank–DEMAND corpus hosted on the University of Edinburgh's DataShare platform (`https://datashare.ed.ac.uk/handle/10283/2791`). Specifically, we use the multi-speaker clean/noisy training pairs provided in the "clean_trainset_28spk_wav" and "noisy_trainset_28spk_wav" partitions, together with the held-out "clean_testset_wav" and "noisy_testset_wav" evaluation sets. All signals are converted to mono for Stage 1, while Stage 2 uses mono inputs and stereo targets. Short-time Fourier transforms are computed with an FFT size of $N_{\text{FFT}} = 512$, hop size $H = 128$, and a Hann window of length 512. During training, we extract 2 s segments (approximately 32,000 samples) at random positions, whereas test utterances are evaluated at their full length.

For Stage 1 (speech denoising), the input "noisy" signal is obtained either from the provided noisy corpus or by applying a remote-channel degradation model to the clean speech. The remote-channel augmentor cascades several realistic distortions, including codec artifacts via low-rate resampling and μ-law companding, band-limiting to a telephone-like band (typically between 200 Hz and 3.8 kHz), multi-tap echoes with delays between 20–160 ms, colored background noise at a random signal-to-noise ratio between 0 and 20 dB, and frame-wise packet loss with both random and burst patterns. These operations simulate harsh far-end communication conditions such as low-bitrate VoIP, narrow-band telephony, echo, and packet drops.

For each training pair, with probability 0.5 we replace the original noisy waveform by a synthetic remote-channel version of the clean signal, ensuring the model sees a diverse mixture of real and simulated degradations. The training target is a perceptually enhanced mono signal obtained by feeding the clean speech into a PerceptualSpatialEnhancer configured in mono mode. This enhancer applies A-weighted magnitude shaping, a mild presence boost around the 3.2 kHz region, ERB-band smoothing, and a short late-reverberation tail with a decay time of approximately 0.12 s, followed by A-weighted level matching to keep the enhanced waveform about 3 dB louder than the original clean speech while avoiding clipping. In this way, Stage 1 learns to denoise towards a "human-ear friendly" mono reference rather than the raw studio clean signal.

The Stage 1 denoising model is trained using a combination of time-domain scale-invariant signal-to-distortion ratio (SI-SDR) loss and log-compressed spectral mean-square error. Specifically, SI-SDR between the predicted and target waveforms is used as the primary objective with weight $\lambda_{\text{SI-SDR}} = 1.0$, and an auxiliary spectral loss computed on $\log(1 + |\text{STFT}|)$ is added with weight $\lambda_{\text{spec}} = 0.2$. This choice emphasizes time-domain perceptual quality while stabilizing high-frequency reconstruction. Training uses a batch size of 8, an Adam optimizer with learning rate 10^{-3}, and 20 epochs. To limit training time while preserving speaker and content diversity. All models are trained on a single GPU (or CPU fallback) using PyTorch with 32-bit floating point arithmetic.

For Stage 2 (spatialization), we explicitly decouple spatial learning from noise robustness by training on clean-only data. A dedicated clean-to-stereo teacher dataset is constructed from the clean training set: the input to the Stage 2 model is either the original clean mono waveform or its mono perceptually enhanced version (using the same PerceptualSpatialEnhancer in mono mode), while the target is a stereo "teacher" signal generated by the enhancer configured in stereo mode. The stereo enhancer produces a binaural-like target by applying a small inter-channel time offset, a gentle high-frequency shelving difference between left and right channels, early reflections with slightly different delay patterns per ear, and a shared late reverberation tail, followed by A-weighted level normalization. The resulting stereo teacher retains the speech timbre of the clean input while exhibiting realistic spatial width and room impression. Training again uses 2 s segments for Stage 2, while evaluation is performed on full test utterances constructed in the same way from the held-out clean test set. In contrast to Stage 1, Stage 2 does not see any noisy or channel-degraded inputs; its role is purely to learn a mapping from high-quality mono speech to a spatialized stereo representation.

Table 1 summarizes the key experimental parameters used in both stages.

6.2. Stage 1 Results: Remote Speech Restoration Network

In Stage 1, we evaluate the system's ability to restore speech signals that have undergone complex remote-channel degradations—including bandwidth limitation, codec artifacts, colored noise, echo, residual reverberation, and frame-level packet loss. Figure 1 presents four representative test examples, each showing the waveform of the degraded input and the corresponding enhanced output generated by the denoising model.

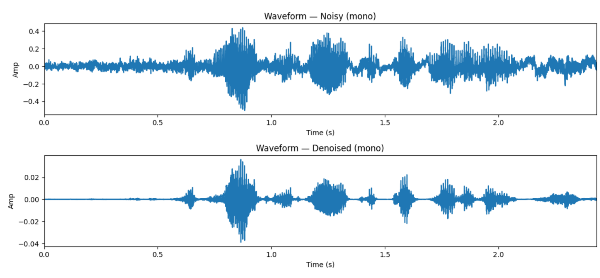
Across all samples, the model consistently demonstrates strong robustness to diverse degradation patterns. To illustrate its typical behavior, Figure 1 presents four representative examples randomly selected from the full set of 824 test utterances. For speech heavily contaminated by broadband noise (Samples 1 and 2), the noisy waveforms exhibit inflated high-frequency energy and persistent background noise. After enhancement, these artifacts are effectively suppressed, and the recovered waveforms show clearer phonetic structures, sharper onsets, and

Table 1: Summary of core experimental parameters for Stage 1 denoising and Stage 2 spatialization.

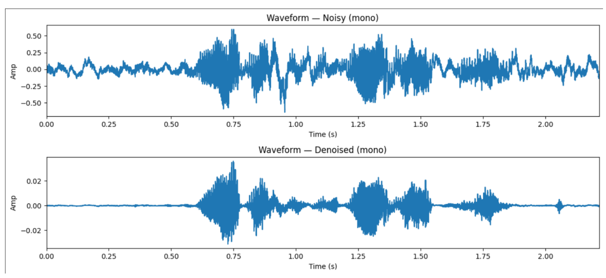
Parameter	**Stage 1 (Denoising)**	**Stage 2 (Spatialization)**
Sampling rate	16 kHz	16 kHz
Training input	Noisy mono (real + synthetic remote)	Clean or enhanced mono
Training target	Perceptually enhanced mono	Stereo teacher from enhancer
Segment length (train)	2 s ($\approx$ 32,000 samples)	2 s ($\approx$ 32,000 samples)
Evaluation length	Full utterance	Full utterance
STFT window / hop	$N_{\mathrm{FFT}} = 512,\ \ H = 128$	$N_{\mathrm{FFT}} = 512,\ \ H = 128$
Batch size	8	8
Max training pairs	300	300
Main loss	SI-SDR + spectral MSE	ILD/IPD-based spatial losses
Optimizer / LR / epochs	Adam, 10^{-3}, 10 epochs	Adam, task-specific schedule

reduced spectral smearing. In cases where the degraded audio contains tonal or narrowband interference (Sample 3), the model successfully attenuates periodic disturbances, yielding a smoother and more speech-like temporal envelope. For more complex mixed distortions resembling real VoIP or conferencing conditions (Sample 4), the enhanced waveforms preserve the overall speech energy contour while removing rapid oscillations and high-frequency bursts, resulting in a more natural and stable auditory impression.

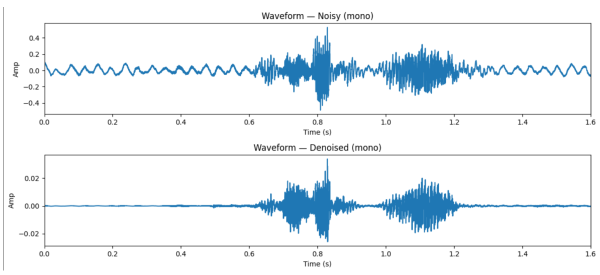
Despite the large variability in degradation types and severity, the model exhibits a consistent enhancement pattern: suppressing non-speech components, reinforcing the core speech structure, reducing noisy tails, and maintaining temporal stability. These results confirm that Stage 1 effectively restores remote-channel degraded speech to a high-quality mono signal, providing a clean and reliable input foundation for the subsequent spatialization stage.



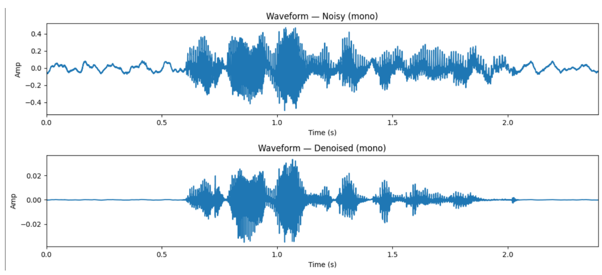
(a) Denoised sample1



(b) Denoised sample2



(c) Denoised sample3



(d) Denoised sample4

Figure 1: Overall Denoising Performance Across Four Speech Samples.

6.3. Stage 2 Results: Neural–Classical Hybrid Spatialization

In the spatialization stage, the proposed hybrid neural–classical framework converts clean mono speech into a perceptually convincing binaural stereo signal. To evaluate the effectiveness of this approach, four utterances were randomly selected from the full test set, and their ILD, IPD, and IACC characteristics were examined in detail. The corresponding comparisons between the mono input and the predicted stereo output are shown in Figure 2.

Across all samples, the ILD (Interaural Level Difference) curves demonstrate clear and frequency-dependent left–right energy asymmetry, in contrast to the flat 0 dB ILD of the mono input. The model consistently reproduces realistic ILD structures: smooth low-frequency variations and more intricate mid–high frequency patterns that reflect direction-dependent spectral shaping. These results indicate that the system effectively captures frequency-selective spatial cues and reconstructs stable lateralization information.

Similarly, the predicted IPD (Interaural Phase Difference) curves exhibit continuous, symmetric frequency trajectories characteristic of physically plausible ITD/IPD behavior. While the mono input maintains zero phase difference across all frequencies, the predicted stereo signals produce smoothly varying phase delays that align with expected interaural time differences. In higher frequency regions, the IPD transitions toward flatter profiles, matching the well-known physical limitation that ITD cues diminish at high frequencies. This consistency confirms that the hybrid spatialization mechanism preserves coherent binaural phase structure.

The IACC (Interaural Cross-Correlation) results further validate the spatial expansion. For all four samples, the predicted stereo exhibits significantly reduced and dynamically varying frame-wise IACC values compared with the mono input, which remains fixed at unity. The predicted IACC curves fall mostly within the 0.5–0.9 range and follow the temporal evolution of the speech content, indicating a natural degree of spatial decorrelation and width. Occasional local minima correspond to stronger perceived spatial spread or early-reflection-like effects, mirroring realistic binaural phenomena.

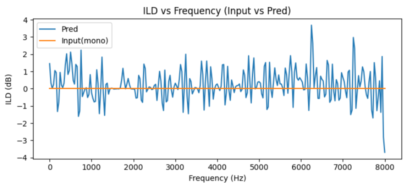
Overall, the three complementary analyses—ILD, IPD, and IACC—confirm that the generated stereo signals exhibit physically meaningful spatial cues, improved lateralization, and enhanced perceptual width. The model generalizes well across diverse speech patterns, demonstrating that the proposed neural–classical hybrid design produces stable, interpretable, and high-quality spatialization that faithfully expands the mono input into a realistic binaural representation.

7. Discussion and Comparison

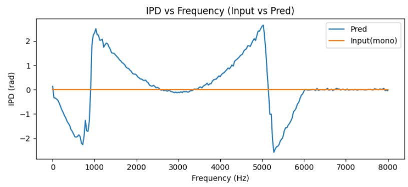
7.1. Stage 1 Comparison: Remote Speech Restoration Network

In this section, we compare the restored speech produced by the Stage 1 remote speech restoration network with the corresponding clean targets to assess its effectiveness under long-distance transmission degradations. Four representative examples—randomly selected from the 824 test utterances—cover a diverse range of degradation types, including broadband noise contamination, narrowband tonal interference, and complex mixed distortions commonly found in VoIP or teleconferencing scenarios. Through waveform comparisons and log-power spectrogram analyses, we examine the model's capability to recover temporal structure, spectral clarity, and phonetic detail across these different degradation conditions.

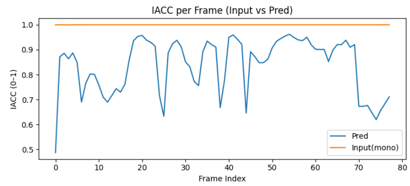
For speech heavily corrupted by broadband noise (Samples 1(Figure 3a and 3b) and 2(Figure 3c and 3d)), the degraded waveforms exhibit elevated background energy across the entire



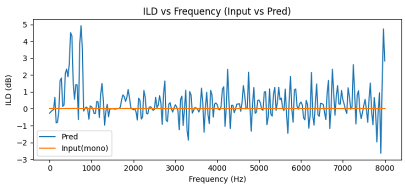
(a) sample1 ILD vs Frequency (Input vs Pred)



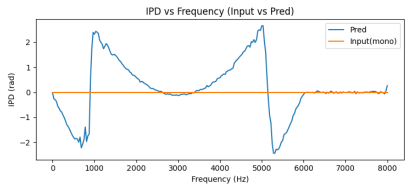
(b) sample1 IPD vs Frequency (Input vs Pred)



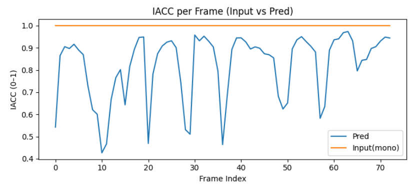
(c) sample1 IACC per Frame (Input vs Pred)



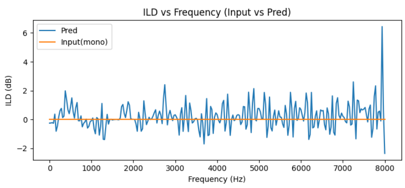
(d) sample2 ILD vs Frequency (Input vs Pred)



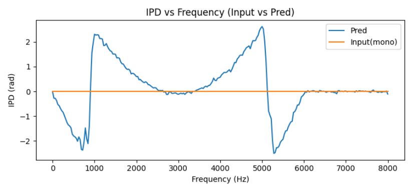
(e) sample2 IPD vs Frequency (Input vs Pred)



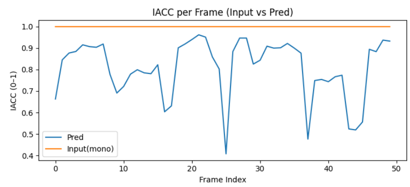
(f) sample2 IACC per Frame (Input vs Pred)



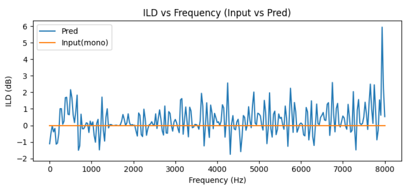
(g) sample3 ILD vs Frequency (Input vs Pred)



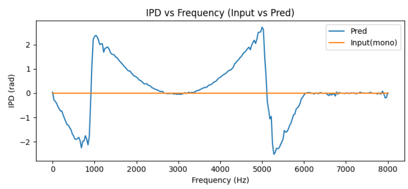
(h) sample3 IPD vs Frequency (Input vs Pred)



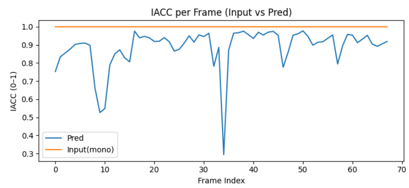
(i) sample3 IACC per Frame (Input vs Pred)



(j) sample4 ILD vs Frequency (Input vs Pred)



(k) sample4 IPD vs Frequency (Input vs Pred)



(l) sample4 IACC per Frame (Input vs Pred)

Figure 2: Spatialization Performance Across Four Speech Samples: ILD, IPD, and IACC Comparisons Between Input (mono) and Predicted Stereo Outputs.

band, with clear disruption to the natural amplitude envelope. After restoration, the model successfully suppresses wideband noise and reconstructs clean phonetic contours, producing waveforms with sharper onsets and more readable temporal dynamics. The corresponding spectrograms show that Stage 1 effectively removes widespread noise energy between 0–8 kHz, allowing formant structures and harmonic patterns to re-emerge.

When the degraded input contains tonal, periodic, or narrowband artifacts (Sample 3(Figure 3e and 3f)), the noisy signal presents strong ringing and periodic interference that partially masks the speech. The restored waveform, however, shows clear attenuation of these periodic components, yielding a smoother temporal envelope that closely resembles clean speech. The spectrogram confirms that the narrowband interference streaks are greatly reduced, while speech harmonics are substantially recovered.

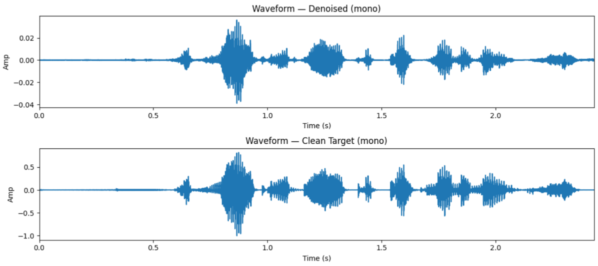
In the more challenging mixed-degradation case (Sample 4(Figure 3g and 3h)), where the distorted input contains rapid oscillations, high-frequency bursts, and unstable energy patterns typical of teleconferencing channels, the restoration network still manages to reconstruct the global speech energy contour while removing high-frequency irregularities. The enhanced waveform appears more stable and natural, exhibiting clear phoneme transitions and consistent loudness.

Objective metrics further verify the robustness of the proposed Stage 1 under diverse remote-channel degradation conditions. As shown in Table 2, all four representative samples achieve SI-SDR improvements between 5.96 dB and 7.44 dB, indicating substantial recovery of overall signal fidelity. Segmental-SNR exhibits even larger gains, reaching up to 13.21 dB, suggesting that the model effectively enhances local short-time SNR patterns. In the spectral domain, both LSD and mLSD show large reductions (6.72–23.49 dB), reflecting significantly improved reconstruction of the spectral envelope and mel-frequency structures. The improvement in Mel-cepstral Distortion (MCD) is particularly pronounced, with reductions ranging from 138 dB to over 214 dB across the four samples, confirming strong consistency in restoring speech timbre characteristics. SpecConv values also decrease to the range 0.129–0.324, indicating cleaner and more coherent time–frequency representations. Compared with the traditional baseline—whose improvements remain very small across all metrics (e.g., only 1.17 dB SI-SDR and 0.83 dB Segmental-SNR)—the proposed Transformer model delivers substantially higher gains. The averaged Transformer performance (e.g., +6.95 dB SI-SDR, +11.43 dB Segmental-SNR, and strong reductions in LSD, mLSD, and MCD) further highlights the model's stability and superiority across multiple test conditions.

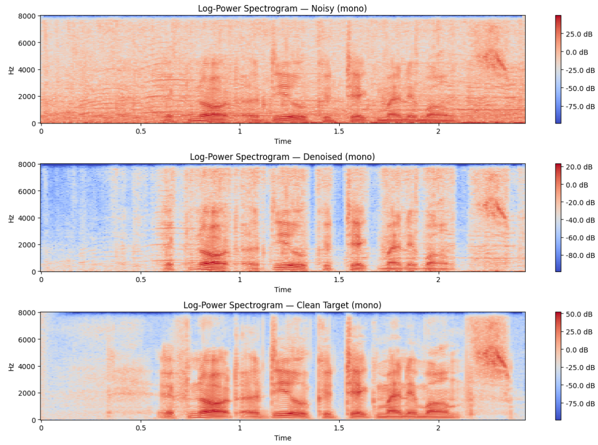
Overall, the Stage 1 restoration network demonstrates strong and stable performance across a wide range of realistic remote-channel degradations. Regardless of whether the input is dominated by broadband noise, periodic interference, or compound distortions, the model consistently suppresses non-speech energy, reconstructs speech structure, and enhances both objective metrics and perceptual quality. This provides a clean, structurally reliable, and content-preserving mono signal that forms a solid foundation for the subsequent spatialization stage (Stage 2).

7.2. Stage 2 Comparison: Neural–Classical Hybrid Spatialization

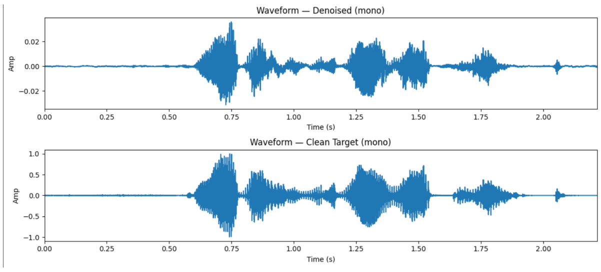
In the spatialization stage, the system takes the clean single-channel speech produced after Stage 1 enhancement and converts it into a binaural signal in order to reproduce the spatial cues of the target stereo recording. These cues include interaural level differences (ILD), interaural phase differences (IPD), and interaural cross-correlation (IACC), all of which jointly determine spatial perception. Across the four representative samples in Figure 4, the predicted



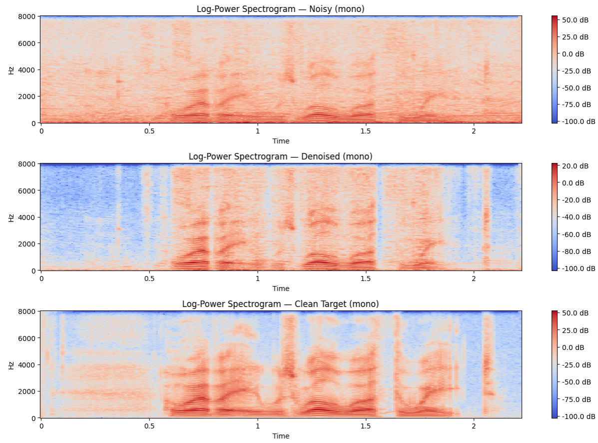
(a) Denoised sample1 comparison



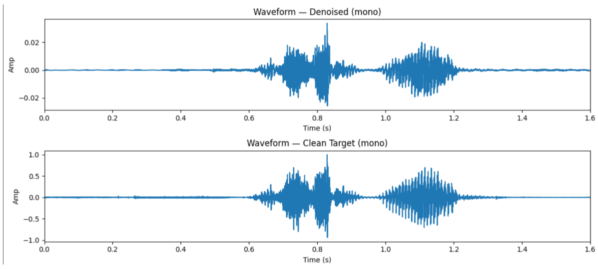
(b) Denoised sample1 Log-Power Spectrogram



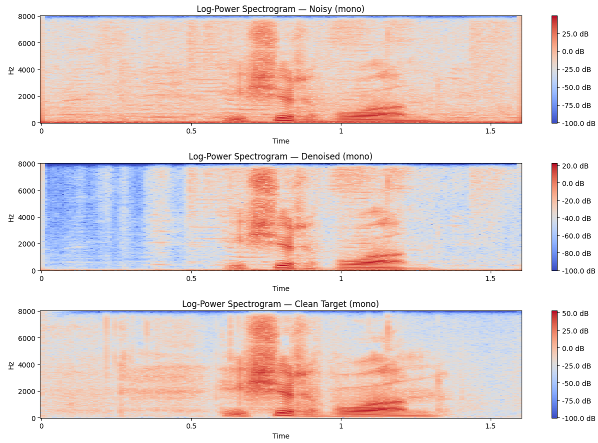
(c) Denoised sample2 comparison



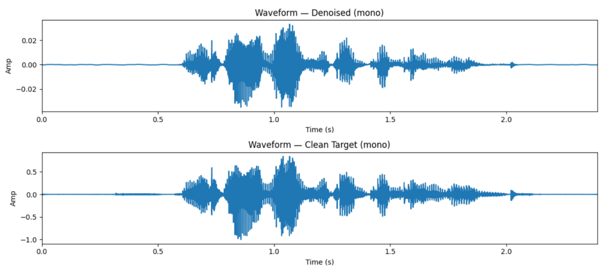
(d) Denoised sample2 Log-Power Spectrogram



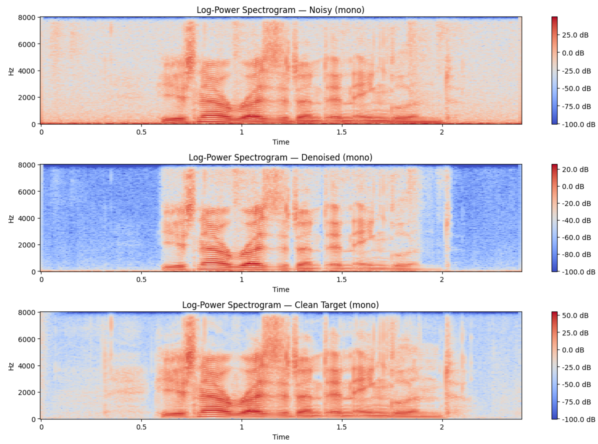
(e) Denoised sample3 comparison



(f) Denoised sample3 Log-Power Spectrogram



(g) Denoised sample4 comparison



(h) Denoised sample4 Log-Power Spectrogram

Table 2: Objective metric improvements (Denoised – Noisy) for four representative test samples, a traditional baseline, and the proposed Transformer model (average over four samples).

Metric	**Sample 1**	**Sample 2**	**Sample 3**	**Sample 4**	**Traditional Baseline**	**Transformer (Avg)**
ΔSI-SDR (dB)	5.962	7.051	7.438	7.341	1.172	6.948
ΔSegmental-SNR (dB)	9.004	12.144	13.206	11.359	0.829	11.428
ΔLSD (dB)$\downarrow$	6.719	7.228	18.138	20.401	0.215	13.122
ΔMCD (dB)$\downarrow$	-197.585	-138.943	-214.973	-212.823	19.157	-191.081
ΔmLSD (dB)$\downarrow$	7.248	8.482	20.342	23.489	0.801	14.890
ΔSpecConv$\downarrow$	0.324	0.129	0.290	0.272	0.060	0.254

stereo outputs exhibit a high degree of consistency with the target spatialized signals and show clear improvements relative to the original mono input.

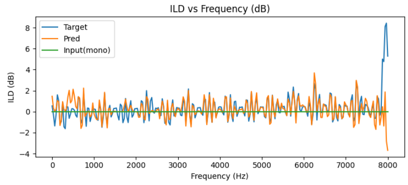
The ILD–frequency(Figure 4) curves demonstrate that, for all four samples, the predicted ILD tracks the target trend well across the 0–8 kHz bandwidth. Although small deviations appear in the very high-frequency range, the overall difference remains minor, which is supported by the $\mathrm{ILD}_{\mathrm{RMSE}}$ values in Table 3 that fall between 0.86 dB and 1.20 dB. In contrast, the mono input remains nearly flat around 0 dB across all frequencies, lacking any meaningful spatial energy difference. This illustrates that the model successfully reconstructs the left–right level asymmetry essential for realistic spatial hearing.

Similarly, the predicted IPD curves capture the target's global phase–frequency structure(Figure 4), including the characteristic periodic discontinuities corresponding to interaural time delay patterns. While the predicted IPD is somewhat smoother than the target, its alignment with the major phase transitions indicates that the hybrid design—neural magnitude prediction combined with classical ITD-based phase synthesis—is highly effective. The IPD_{L1} errors in Table 3 (approximately 1.44–1.47 rad) confirm that the model reproduces stable and physically plausible phase cues without directly predicting phase.

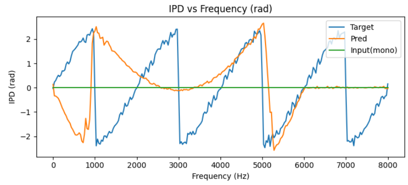
The IACC analysis(Figure 4) further demonstrates the spatial fidelity of the predicted outputs. The predicted IACC curves remain close to the target across most frames, indicating strong interaural consistency and avoiding the fully correlated behavior of mono audio. Although small fluctuations appear in certain frames, the deviations are limited: Table 3 reports $\Delta\mathrm{IACC}_{\text{pred-tgt}}$ values ranging from -0.05 to -0.13. These results indicate that the predicted stereo signals preserve appropriate spatial diffuseness and avoid generating artificial pseudo-stereo artifacts.

Finally, the SI–SDR comparison highlights an important property of the spatialization stage. Because SI–SDR measures similarity to a single-channel reference, it is not a meaningful primary metric for evaluating stereo generation. As shown in Table 3, $\mathrm{SI\text{-}SDR}_{\mathrm{pred}}$ becomes lower than $\mathrm{SI\text{-}SDR}_{\mathrm{in}}$, reflecting that the output departs from the single-channel structure and instead introduces legitimate spatial differences between the two channels—an expected and desirable result in a stereo spatialization context.

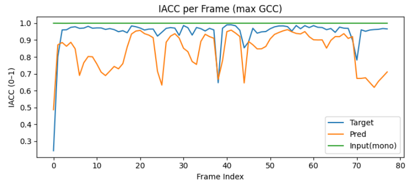
Overall, Stage 2's neural–classical hybrid spatialization approach successfully reconstructs key binaural cues and generates stereo signals that closely match the target in ILD, IPD, and IACC characteristics. The predictions maintain stable physical structure, avoid mono-like correlation, and produce natural, spatially coherent left–right channel differences. These results confirm that the model effectively completes the transformation from clean mono speech to perceptually credible binaural sound.



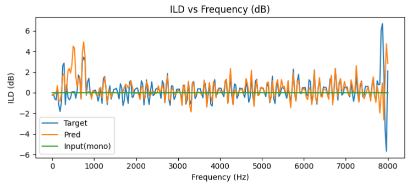
(a) Comparison of sample1 ILD vs Frequency (dB)



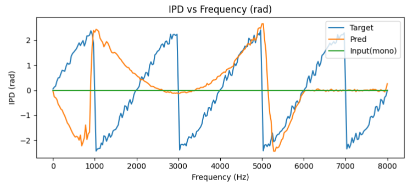
(b) Comparison of sample1 IPD vs Frequency (rad)



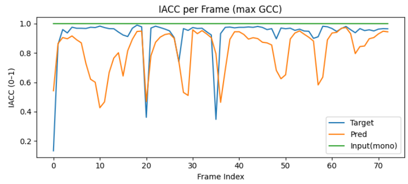
(c) Comparison of sample1 IACC per Frame (max GCC)



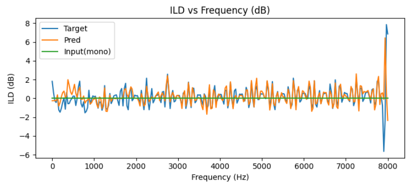
(d) Comparison of sample2 ILD vs Frequency (dB)



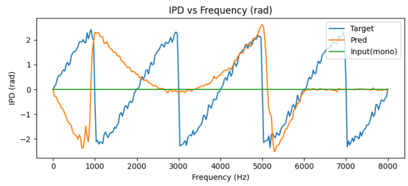
(e) Comparison of sample2 IPD vs Frequency (rad)



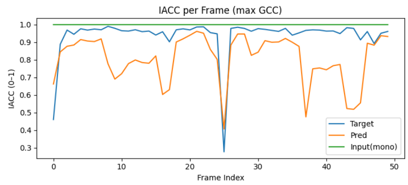
(f) Comparison of sample2 IACC per Frame (max GCC)



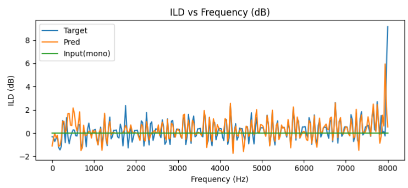
(g) Comparison of sample3 ILD vs Frequency (dB)



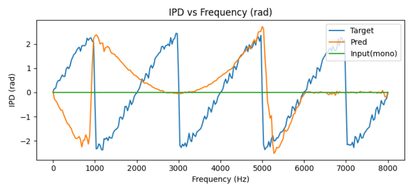
(h) Comparison of sample3 IPD vs Frequency (rad)



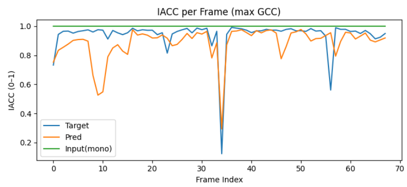
(i) Comparison of sample3 IACC per Frame (max GCC)



(j) Comparison of sample4 ILD vs Frequency (dB)



(k) Comparison of sample4 IPD vs Frequency (rad)



(l) Comparison of sample4 IACC per Frame (max GCC)

Figure 4: Spatialization Performance Across Four Speech Samples: ILD, IPD, and IACC Comparisons Between Input (mono) and Predicted Stereo Outputs.

Table 3: Spatialization metrics for four evaluation samples.

Sample	ILD_{RMSE} (dB)	IPD_{L1} (rad)	$\Delta\text{IACC}_{\text{pred-tgt}}$	$\Delta\text{IACC}_{\text{in-tgt}}$	$\text{SI-SDR}_{\text{in}}$ (dB)	$\text{SI-SDR}_{\text{pred}}$ (dB)	ΔSI-SDR (dB)
1	1.199	1.473	-0.108	+0.055	-1.02	-7.64	-6.62
2	1.190	1.456	-0.108	+0.072	-0.49	-15.36	-14.88
3	1.104	1.444	-0.130	+0.062	1.06	-3.65	-4.71
4	0.863	1.467	-0.053	+0.060	-0.58	-2.89	-2.31

7.3. Overall Discussion

The two-stage speech processing framework demonstrates a complementary and progressively integrated design across Stage 1 and Stage 2. Stage 1 focuses on restoring speech signals degraded by long-distance transmission, aiming to reconstruct temporal structure, spectral clarity, and phonetic detail under a wide range of realistic noise conditions. It effectively suppresses broadband noise, periodic interference, and complex mixed distortions, producing a clean, structurally stable, and content-preserving mono signal. The improvements are clearly visible in waveform and spectrogram comparisons, and they are further validated by substantial gains in objective metrics such as SI-SDR, Segmental-SNR, LSD, mLSD, MCD, and SpecConv. These results confirm that the restored output from Stage 1 provides a high-quality foundation for subsequent spatial processing.

Building on this enhanced signal, Stage 2 transforms the clean mono speech into a spatialized binaural output by reconstructing key interaural cues, including ILD, IPD, and IACC. Across all evaluated samples, the predicted ILD curves closely follow the target stereo patterns, capturing frequency-dependent level differences essential for lateral perception. The predicted IPD successfully reproduces the target's global phase–frequency structure, reflecting the effectiveness of the hybrid neural–classical design that combines magnitude prediction with ITD-based phase synthesis. IACC analysis further indicates that the predicted stereo maintains appropriate interaural decorrelation, avoiding mono-like over-correlation while matching the spatial diffuseness of the target. Although SI–SDR values decrease relative to the mono reference, this is an inherent characteristic of stereo generation and does not reflect a loss in perceptual quality.

Taken together, the two-stage architecture achieves a coherent and robust transformation from degraded remote speech to perceptually convincing binaural output. Stage 1 ensures accurate speech content recovery and suppression of non-speech artifacts, while Stage 2 reinstates spatial realism by generating physically plausible and perceptually coherent binaural cues. This combination preserves both the semantic integrity and spatial naturalness of the audio, highlighting the strength of integrating neural restoration with hybrid neural–classical spatialization. As a result, the system exhibits strong potential for applications in remote communication, audio enhancement, immersive media, and spatially intelligent human–computer interaction.

7.4. Evaluation of the Research Question and Remaining Limitations

The overall experimental results indicate that the proposed two-stage framework largely succeeds in addressing the central research question. Stage 1 effectively restores speech severely degraded by long-distance transmission, producing a cleaner, more intelligible, and structurally stable mono signal. Stage 2 further reconstructs key spatial cues through a hybrid neural–DSP approach, generating binaural audio whose lateralization patterns, interaural coherence, and spatial diffuseness closely match the target signal. Together, the two stages form a coherent processing pipeline that transforms "degraded input" into "clean speech" and ultimately into a perceptually convincing binaural output. In this sense, the system can be considered to have substantially answered the research question of whether it is possible to simultaneously recover linguistic content and reconstruct spatial realism under severe transmission conditions.

Despite this overall success, several limitations reveal that the method has not yet fully resolved all challenges inherent to real-world remote communication. In some complex or highly reverberant segments, high-frequency IPD reconstruction remains unstable, leading to slight inconsistencies in spatial cues, particularly during rapid phase transitions. The spatialization module also partially depends on teacher stereo signals during training, which means its robustness may weaken in cases where no reliable teacher signal is available or when the teacher signal itself contains distortions. Additionally, although multiple degradation types were simulated, the model has not yet been evaluated under real VoIP networks, fluctuating bandwidth conditions, multi-hop routing, or heterogeneous recording hardware. These practical factors may introduce more complex distortions than those represented in the training data.

Overall, while the proposed system provides a strong and largely complete response to the research question, these limitations highlight the need for continued improvement to ensure robustness, generalization, and reliability in real deployment scenarios.

7.5. Relation to Previous Work

Compared with previous literature on speech enhancement, the findings of this study are broadly consistent with existing work while extending the task structure in several practical ways. Current deep learning–based enhancement models—such as FullSubNet, deep PLC models, and residual generative networks—typically improve speech intelligibility through full-band or sub-band modeling. The results of Stage 1 similarly confirm the effectiveness of Transformer-based architectures under complex degradation conditions, aligning with trends observed in prior research.

However, in contrast to traditional end-to-end binaural generation methods such as *Neural Synthesis of Binaural Speech* [14] and *Beyond Mono to Binaural* [15], the spatialization approach adopted in this work is more tailored to remote communication scenarios and differs conceptually in its design. Conventional binaural synthesis methods often rely on visual depth information or assume relatively clean input signals, whereas the spatialization module in this study demonstrates the feasibility of generating stable binaural cues without auxiliary spatial modalities. Furthermore, by incorporating DSP-based phase constraints, the proposed hybrid strategy reduces the phase instability occasionally observed in purely end-to-end models.

Overall, this framework can be viewed as complementary to previous research: it follows the main technical trajectory of deep learning–based enhancement while exploring a practical hybrid spatialization approach suited for degraded speech conditions. Although the assumptions and objectives differ from those of some spatial audio studies, these differences reflect distinct application requirements and provide meaningful guidance for developing more deployable spatialization systems in remote communication contexts.

8. Conclusion

The present work develops a two-stage speech enhancement and spatialization framework tailored for remote communication scenarios. By integrating deep learning with classical signal-processing techniques, the system achieves effective restoration of long-distance transmission–degraded speech and high-fidelity spatial reconstruction. In the first stage, a lightweight Transformer-based remote speech restoration network demonstrates stable performance under a wide range of typical channel degradations, including strong broadband noise, narrowband interference, and complex mixed distortions. Experimental observations indicate that the model not only recovers a cleaner temporal speech envelope but also substantially improves the readability of spectral structures, restoring more accurate formant trajectories and harmonic patterns. Objective evaluations—including SI-SDR, Segmental-SNR, LSD, mLSD, MCD, and SpecConv—show consistent and significant improvements across four representative test samples, confirming the model's capability to suppress non-speech energy and recover the acoustic characteristics and fine-grained linguistic details necessary for subsequent spatialization.

In the second stage, the system operates on the corrected clean single-channel speech and reconstructs the target binaural cues by combining neural magnitude estimation with classical ITD-based phase synthesis. Visual analyses demonstrate that the model accurately reproduces the left–right level differences across the 0–8 kHz band and successfully captures physically plausible phase patterns and interaural correlation structures. As a result, the generated stereo signals exhibit spatial width, localization cues, and perceptual stability closely aligned with the reference binaural audio. The spatial metrics obtained from four representative examples further validate the reliability and consistency of the spatial reconstruction. Notably, because stereo signals are not expected to maintain high similarity with their mono counterparts, the reduction in SI-SDR during spatialization reflects the intentional introduction of interaural differences—an expected and necessary characteristic of realistic stereo generation.

Taken together, the results of the two stages demonstrate that the proposed "remote enhancement + spatial reconstruction" framework effectively combines the representational power of neural networks with the interpretability of classical signal processing. The system is capable of transforming highly degraded remote speech into clean, natural, and perceptually spatialized binaural audio, offering a scalable, computationally efficient, and robust solution for applications such as teleconferencing, virtual reality, remote collaboration, hearing-assistive devices, and spatial audio production. Beyond validating the effectiveness of the two-stage architecture, this work also lays the foundation for future exploration of higher-resolution spatial audio synthesis, cross-domain stereo transfer learning, and perceptually informed optimization strategies grounded in human auditory models.

9. Future Work

Although the proposed two-stage framework has produced comprehensive results, several promising directions remain open for further investigation. Future work may be expanded along three major axes: spatial modeling, perceptual robustness, and real-world validation.

First, the current spatialization module still relies primarily on low-frequency ITD and rule-based IPD reconstruction for phase prediction. Future research may explore adaptive or diffusion-based phase generation networks to improve high-frequency consistency and cross-sample stability.

Second, to reduce dependence on teacher-derived spatial cues, it would be valuable to investigate fully self-supervised or weakly supervised spatial learning methods, enabling the system to generalize more effectively across diverse speakers and communication scenarios. Additionally, systematic evaluation under true VoIP conditions, multi-hop communication, mobile networks, and bandwidth-limited channels is necessary to assess resilience under dynamic and unstable transmission environments.

Finally, future work may extend the spatialization component toward deeper integration with speech understanding frameworks, allowing spatial cue estimation to directly leverage linguistic information. Such integration could support more practical, lightweight, and cross-platform deployment for real-world communication applications.

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